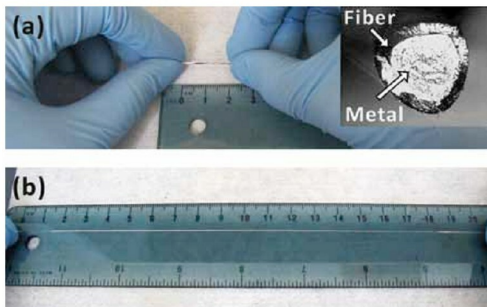




Super-stretchable conduction wire

told us that BCI gives them hope for the future."

Scheuermann's brain was implanted with two electrode grids, which have 96 tiny contact points for brain areas that control right arm and hand movement. The electrodes pick up neurons firing to activate arm movement.

Researchers estimate that similar robot arms could be available to far more patients in five to ten years. The next step for the work will be to include feedback potential in the electrodes so that the brain can interpret sensations like grip strength from the arm.

Fork that talks while you eat

Here comes a fork that can inject a bit of fun and excitement into your lunch if you find mealtimes boring. Researchers from the Ochanomizu University in Tokyo have announced the development of a talking fork called EsThermine that speaks different words and plays bizarre sounds as you eat.

The fork-like instrument comes equipped with a microcontroller, a speaker, amplifier and battery within its handle to produce different sounds (depending on the type of food being eaten) and even speak while you enjoy your meal.

"The sounds are apparently generated by the resistance value of the food as it is bitten off the fork - the noise it produces as one eats a chicken nugget

sounds remarkably similar to a chicken clucking. Apparently, longer pieces of food will cause the fork to produce multiple sounds and words while they are eaten," the Daily Mail reported.

The fork has the ability to distinguish between different types of eating.

With this invention, the developers aim to encourage people to try different types of food and thus a more varied diet. They want the fork to be used as a communication tool for couples, the report added.

Super-stretchable conduction wires for mobile phones

Scientists from North Carolina (NC) State University have come up with new conduction wires that can be stretched up to eight times their original length and used in mobile phones. These wires could be used in headphones as well as phone chargers, and hold potential for use in electronic textiles.

To make the wires, researchers started with a thin tube made of an extremely elastic polymer and then filled the tube with a liquid metal alloy of gallium and indium, which is an efficient conductor of electricity.

"Previous efforts to create stretchable wires focus on embedding metals or other electrical conductors in elastic polymers, but that creates a trade-off," said Dr Michael Dickey, an assistant professor of chemical and biomolecular

engineering at NC State University and co-author of a paper on the research. "Increasing the amount of metal improves the conductivity of the composite but diminishes its elasticity," Dickey said.

"Our approach keeps the materials separate, so you have maximum conductivity without impairing elasticity. In short, our wires are orders of magnitude more stretchable than the most conductive wires, and at least an order of magnitude more conductive than the most stretchable wires currently in the literature," he added.

Bulb powered by gravity

Deciwatt, a part of design company Therefore in London, UK, has developed a technology called GravityLight that allows light bulbs to be powered by gravity.

Wondering how it works? Well, as a part of this technique, a sack of sand is used to gradually pull a piece of rope through a dynamo mechanism, which generates electricity to ultimately power the light-emitting diodes (LEDs). The most startling fact of this technique is that a total of three-second pull on this rope can keep the LED bulb running for up to 30 minutes.

Such applications are viable for countries that do not have a stable and reliable electricity access. The project team plans to distribute these light bulbs for free to billions of poor people who currently rely on biomass fuels like kerosene.

World's first peel-off solar cells

Researchers at Stanford University have created what they claim to be the first flexible, reusable decal-like solar panels that can attach to almost any surface. In other words, it is possible to attach these thin-film solar cells to paper, plastic and glass without direct fabrication. The adhesive comes off with some room-temperature water.

Thin-film solar cells, such as hydrogenated amorphous silicon (a-Si:H), cadmium telluride (CdTe) and copper indium gallium selenide (CIGS), are generally made with rigid silicon and glass structures to maintain their form—according to a paper by lead investigator